

Finding Niche: Platform Cryptocurrency Financial Services and Operations as Support Services for SMEs - Transaction Ledger and Data Storage

July 2025

Tatja Karkkainen, Kiitot, tatja@kiitot.fi

Abstract:

This paper looks at the cost minimisation of storing data on a platform cryptocurrency. On-chain data storage for transaction verification and ledger, as utilised by Solana, can offer significant benefits for SMEs complementary feature provision. By leveraging Solana's capability to handle microtransactions, the study demonstrates transaction traceability and reduced external storage costs. It considers the purposes and requirements to store data for a year, five years or twenty years. Applying economic theories such as network effects, diminishing returns and economies of scale, the paper provides a strategic framework for SMEs to integrate blockchain technology effectively. The findings highlight the significant operational efficiencies and cost savings achievable through blockchain adoption with suggested best practices, while also noting the regulatory hurdles and security concern challenges.

Keywords: Blockchain Technology, SMEs, Financial Services, Data Storage, Economic Theories

JEL Codes: L26, O16, O33

Acknowledgements: Thank you for your comments Dr Michele Fabi, Dr Daniel Broby and Mr Wyatt Clarke.

Introduction

This study is motivated by the current trial-and-error approach observed in blockchain applications for Small and Medium-sized Enterprises (SMEs). SMEs contribute to economies growth, employment and innovation (OECD, 2019). We attempt to approach this by assessing the costs involved using a platform cryptocurrency for SME applied support services. As this a novel area of research, we will also apply relevant theories in blockchain adaptation. For instance, the notion of diminishing returns is important when selecting technologies for it highlights the point at which additional investment yields progressively smaller benefits, helping to avoid over-investing in solutions that provide minimal incremental value. The comprehension of this principle can assist SMEs allocate resources more efficiently, choosing technologies that offer the most significant returns relative to their cost, business models and the size of market opportunity. Whilst blockchain technology continues to evolve with significant applications in various sectors, as demonstrated by Yermack (2017) and novel industries - sharing economies (Hawlitschek et al., 2018) or NFTs (Wang et al. 2021), the rate of blockchain adoption has not matched the initial hype and its investment levels. We will also address a few of the reasons behind the slow adoption by SMEs.

In Table 1, we list common economic theories related to financial technologies and further select notions related specifically to platform cryptocurrency applications in support features such as network effects, platform theory, economies of scale, notion of diminishing returns and game theory. The table also points out whether such a notion can relate to financial services or operations. The aim is to project specific business models, scales of operations and circumstances under which blockchain technology would offer potential solutions for SMEs. Not all companies will scale up to global platforms, in this manner SMEs should consider using third party open platforms to provide some of their service or product features.

The adoption and implementation of blockchain technology in financial services and operations can be better understood through several theoretical lenses. Network effects, as described by Varian & Shapiro (1998) and Metcalfe (2013), highlight how the value of blockchain networks like Solana increases as more participants use the system. This is evident in payment systems and social trading platforms where increased participation enhances utility and efficiency. Also, such incentivise development of more features in an ecosystem. As noted in Table 1, Platform Theory holds that value is created cumulatively through complementary layers built on a shared core (Parker et al., 2016). In effect, a modern platform

functions as an ecosystem, where modular components such as payments, lending, tokens, traceability services can be plugged in or swapped out without disrupting the underlying ledger. Solana, as a platform cryptocurrency, exemplifies this approach: its network offers high-throughput payments, upon which additional modules are continuously added. Third-party services can prosper by plugging into the core ledger, demonstrating the network effects and modularity that make the ecosystem attractive to SMEs. By leveraging this layered structure, smaller firms gain access to sophisticated financial and operational capabilities without the expense of maintaining their own blockchain.

Economies of scale, as posited by Marshall (1920) and Smith (1776), further illustrate how growing blockchain networks can reduce the cost per transaction by spreading fixed costs over a larger number of transactions. This is particularly relevant for SMEs using blockchain for transaction processing and data storage, as the reduced costs can significantly enhance operational efficiency. But conversely, the principle of diminishing returns (Malthus, 1798; Ricardo, 1817) serves as a caution for over-investment in blockchain technology beyond a certain point, where additional resources result in progressively smaller increases in output. This understanding helps in resource allocation, ensuring investments in blockchain yield significant returns relative to costs.

The theory of disruptive innovation, proposed by Christensen (1997, 2015), is also applicable. It posits that smaller companies with fewer resources can challenge established businesses by initially targeting non-premium customers before moving upmarket. Blockchain technology offers a strategic advantage for SMEs to disrupt traditional financial services by providing cost-effective, secure, and efficient transaction solutions.

Regulatory arbitrage, behavioural finance, data network effects, platform theory, technological adoption lifecycle, agency theory, game theory, and peer-to-peer models collectively underpin the strategic framework for integrating blockchain technology in a manner that can be utilised to maximise benefits while mitigating risks.

The structure of the rest paper is as follows, a review of blockchain technology's capabilities, a cost analysis of Solana's data storage, comparative scenarios for rent and rent exemption, discussion of with a set of best practices, recommendations or criticisms for practical implementation, and the final chapter paper concludes

Table 1. Related notions for platform cryptocurrency enabling features in financial services and operations.

Theory / Framework / Notion	Description	Financial Services	Operatio
Network Effects	Value increases as more people use it; seen in payment systems, social trading platforms, and blockchain networks. (Varian & Shapiro, 1998, Metcalfe, 2013)	Yes	Yes
Economies of Scale	Reduced costs per transaction as companies grow due to spread of fixed costs. (Marshall, 1920; Smith, 1776)	Yes	Yes
Diminishing Returns	Adding more resources results in smaller increases in output; leads to inefficiencies beyond a certain point. (Malthus, 1798; Ricardo, 1817).	Yes	Yes
Disruptive Innovation Theory	Smaller companies with fewer resources can challenge established businesses by focusing first non-premium customers and later on premium customers. (Christensen, 1997, 2015)	Yes	Yes
Platform Theory	Connects different stakeholders, benefiting from network effects and scaling rapidly by adding more users/services. (Parker et al. 2016)	Yes	Yes

Regulatory Arbitrage	Exploiting differences in regulations between jurisdictions for competitive advantage. (Houston et al. 2012)	Yes	No
Game Theory	Strategic decision-making in competitive behaviours, pricing strategies, and customer retention. (von Neumann & Morgenstern, 1944).	Yes	No
Behavioural Economics	Psychological influences on financial behaviours; leveraged by fintech apps for user-friendly interfaces. (Kahneman, 2011; Thaler & Sunstein 2008)	Yes	No
Data Network Effects	More data improves products/services, creating a virtuous cycle of better algorithms and customised solutions.	Yes	Yes
Technological Adoption Lifecycle / Innovation Diffusion Theory	Categorises consumers based on their willingness to adopt new technologies, aiding market entry strategies. (Rogers, 2003)	Yes	Yes
Agency Theory	Deals with conflicts of interest between stakeholders, mitigated by transparency and automated contracts. (Jensen & Meckling, 1976).	Yes	Yes
Peer-to-Peer (P2P) Models	Eliminates intermediaries by directly connecting consumers/businesses; seen in lending, payments, and insurance.	Yes	Yes
Information Asymmetry	Reducing information asymmetry leads to more efficient markets through better access to information. (Akerlof, 1970; Stiglitz, 2002).	Yes	Yes

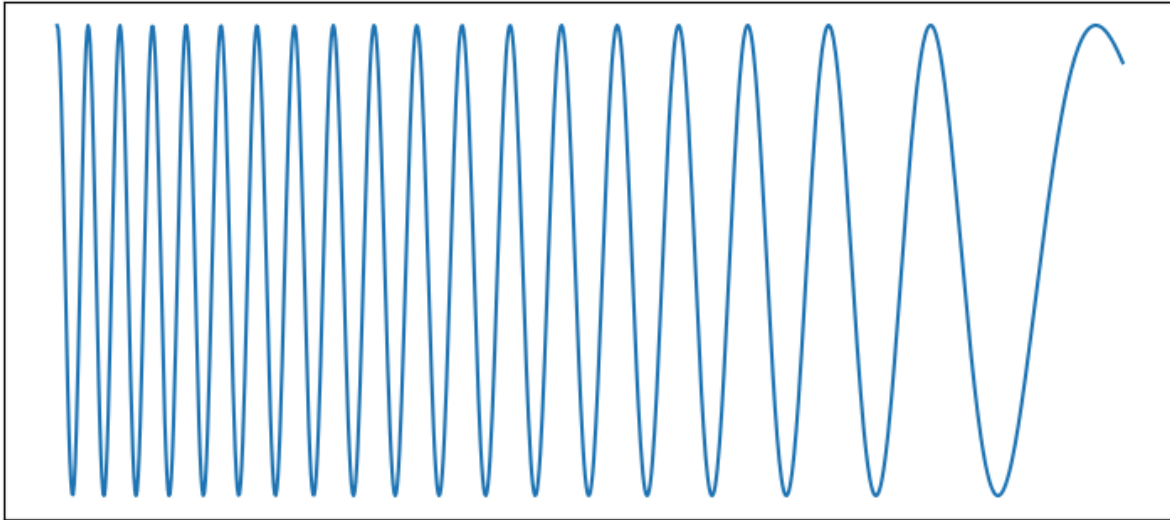
On platform cryptocurrency in operational and financial functionality advantage over traditional banking or in-house built IT operations.

In platform cryptocurrency blockchain platforms such as Solana, Ethereum, Cardano, and Polkadot, the capability to handle minute transaction units is essential for precise financial transactions and detailed record-keeping. On the Solana blockchain, for instance, transaction size can be as low as 0.000000001 Solana coin (SOL), or 1 lamport, showcasing the network's efficiency in managing microtransactions. In this manner the Solana unit fractionality is 10^{-9} . Similarly, Ethereum uses units called wei, where 1 wei equals 10^{-18} ETH, allowing for extremely granular transactions. This precision is particularly important when calculating gas fees for smart contracts, often measured in gwei, which is 10^{-9} ETH. Cardano's smallest unit, known as a lovelace, equals 0.000001 ADA, enabling the blockchain to facilitate detailed transaction handling and microtransactions. Polkadot employs the Planck as its smallest unit, equivalent to 10^{-12} DOT, further illustrating its capacity for high-precision financial operations (Polkadot, 2025). These capabilities highlight the advanced handling of fractional unit transactions by blockchain platforms, in contrast to traditional banking systems, which are typically restricted to two decimal places. Stablecoins can be used for cross-border activities (e.g. Dionysopoulos et al., 2025), although they may be subject to sanctions in certain regions due to regulatory restrictions (WSJ, 2025). Additionally, limited information can be stored on Tether or USDC, as they operate on platform cryptocurrencies such as Ethereum or Solana (e.g. Karkkainen & Broussard, 2025). Our analysis focuses on Solana, due to its specific design for handling microtransactions and its position as the platform cryptocurrency with the highest transaction volume (Coingecko, 2024).

The use of fractions can help for transaction party validation and internal record keeping with accounting, operations including supply chain management. Whilst Solana transactions provide unique transaction signatures, these are nevertheless generated randomly and cannot be pre-requested to identify pre-set purchases.

The platform crypto integrated tools can help in various operational functions that could be considered from payments and log activities to holding issued securities and trading in them. For instance, we could consider the below:

Figure 1. Frequency of transactions from payments to financial security investing



In the high-frequency section of Figure 2, we can place SME money flows, micropayments, and intraday operational logs. Progressing along the spectrum, we encounter simple transactions and supply chain records. At the lower frequencies, we find large money transfers that may relate to the transfer of asset ownership or other financing activities.

In light of this, we should consider not only the transaction fees but also the data rent costs associated with storing transaction records. While payment information may not need to be retained for more than five years, records relating to financial security issuance may require storage for several decades. For example, Article 40(1)(b) of the Fifth Anti-Money Laundering Directive (AMLD5) of the European Union mandates that entities retain transaction records for five years following the end of a business relationship with a customer or the date of an occasional transaction (EBA, 2024).

The computation of the rent optimisation for long term transaction record

On the Solana blockchain, transaction fees are set at a base rate of 5,000 lamports per signature, equivalent to 0.000005 SOL per signature (Solana, 2024A). These fees compensate validators for computational resources and aim to reduce network spam. For example, if the price of SOL is assumed to be \$20, the fee per transaction would be \$0.0001, whereas at \$500 per SOL, the fee would be \$0.0025. Additionally, users may choose to pay optional prioritisation fees to expedite transactions.

This paper focuses specifically on the rent cost associated with transaction data size. The data storage rent is 0.00000348 SOL per byte-year (Solana, 2024A). A typical basic transaction is 267 bytes in size, whereas more complex transactions involving multiple instructions—such as staking or interactions with smart contracts—may require several kilobytes. If a large number of on-chain transactions are to be retained for extended periods, the associated rent fees must be taken into account.

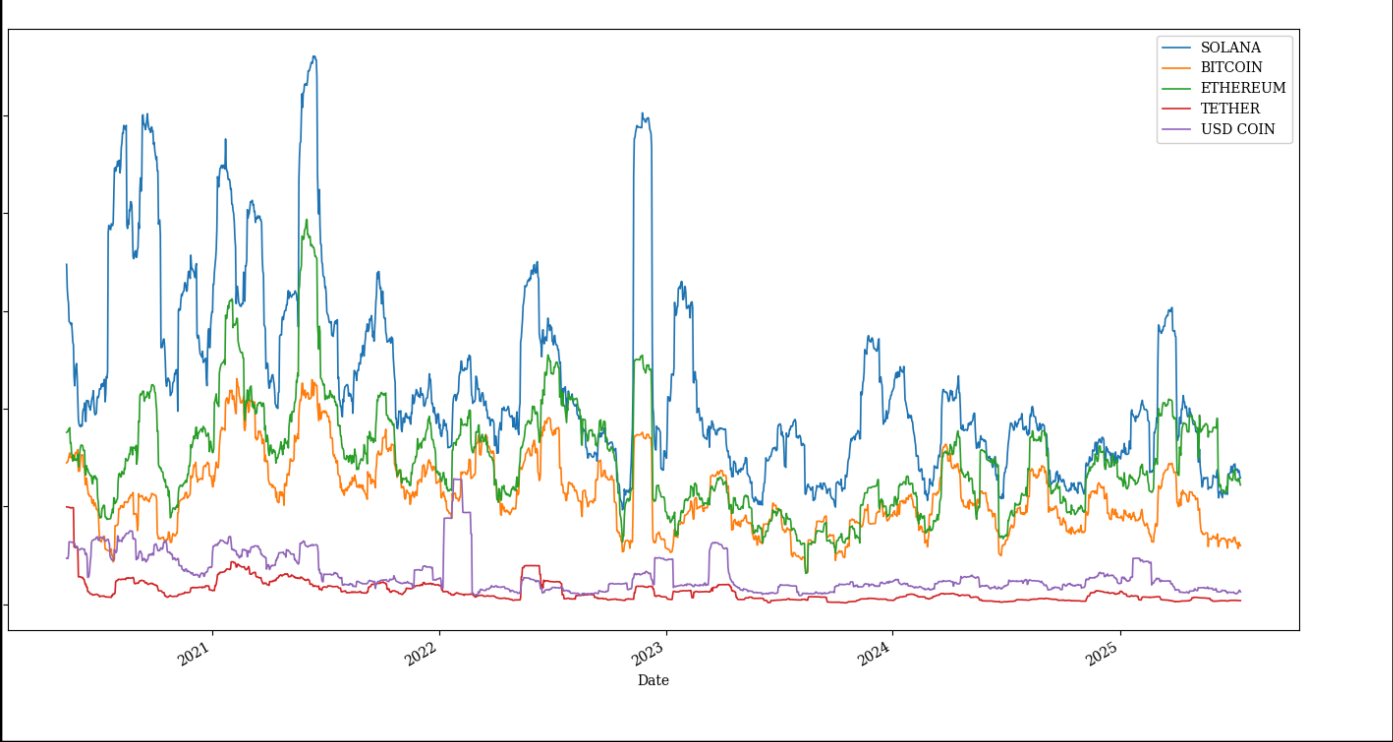
A four-digit operational tag occupies four bytes of on-chain storage. At Solana’s rent rate of 0.00000348 SOL per byte-year, this tag therefore costs

$$4 \times 0.00000348 = 0.00001392 \times 0.00000348 = 0.00001392 \times 0.00000348 = 0.00001392 \text{ SOL per year.}$$

Because Solana produces roughly two blocks per second, each stamped to the nearest second (e.g. “19 May 2023 at 04:56:36”; Solana Explorer, 2024), the network can accommodate 1,393 distinct four-digit identifiers (0000–1392) every second across the two blocks. This capacity is ample for most operational workloads.

Although blockchain technology delivers advantages such as disintermediation through reduced information asymmetry (Davidson et al., 2016) and truly global operational reach (e.g. Di Maggio et al., 2024; Tian, 2017), long-term price volatility remains a key consideration when budgeting for on-chain storage and other recurrent costs. From Figure 2 and Table 3 we could see that Solana is more volatile compared to main cryptocurrencies such as Bitcoin and Ethereum. We can also see the decreasing volatility for the assets represented. This may be a combination of market maturity, positive regulatory environment and favourable macro economic environment and also. That said, it is notable that Solana is significantly more volatile than the S&P 500, a standard proxy for equity markets, yet since its launch it has displayed virtually no correlation with the index as can be seen in Figure 3.

Figure 2. 30-Day Annualised Volatility: SOL, BTC, ETH, USDT, USDC



The 30-day annualised volatility for Solana, Bitcoin, and Ethereum is calculated using daily log returns and a 365-day year. For the stablecoins Tether and USD Coin, volatility is based on daily arithmetic growth rates of market capitalisation. Data are sourced from CoinGecko and cover the period from 11 June 2020 to 14 July 2025. <https://www.coingecko.com/>

Table 2. Annualised Returns and Volatility

	Annualized Return (%)	Annualized Volatility (%)
Solana	97.45	125.16
Bitcoin	54.25	59.53
Ethereum	55.83	79.01
Tether	-0.03	3.88
USD Coin	0.00	4.40

Notes: Daily log returns and standard deviations for Solana, Bitcoin, and Ethereum are calculated and subsequently annualised using a 365-day calendar year. Data are sourced from CoinGecko and cover the period from 11 June 2020 to 14 July 2025. <https://www.coingecko.com/>

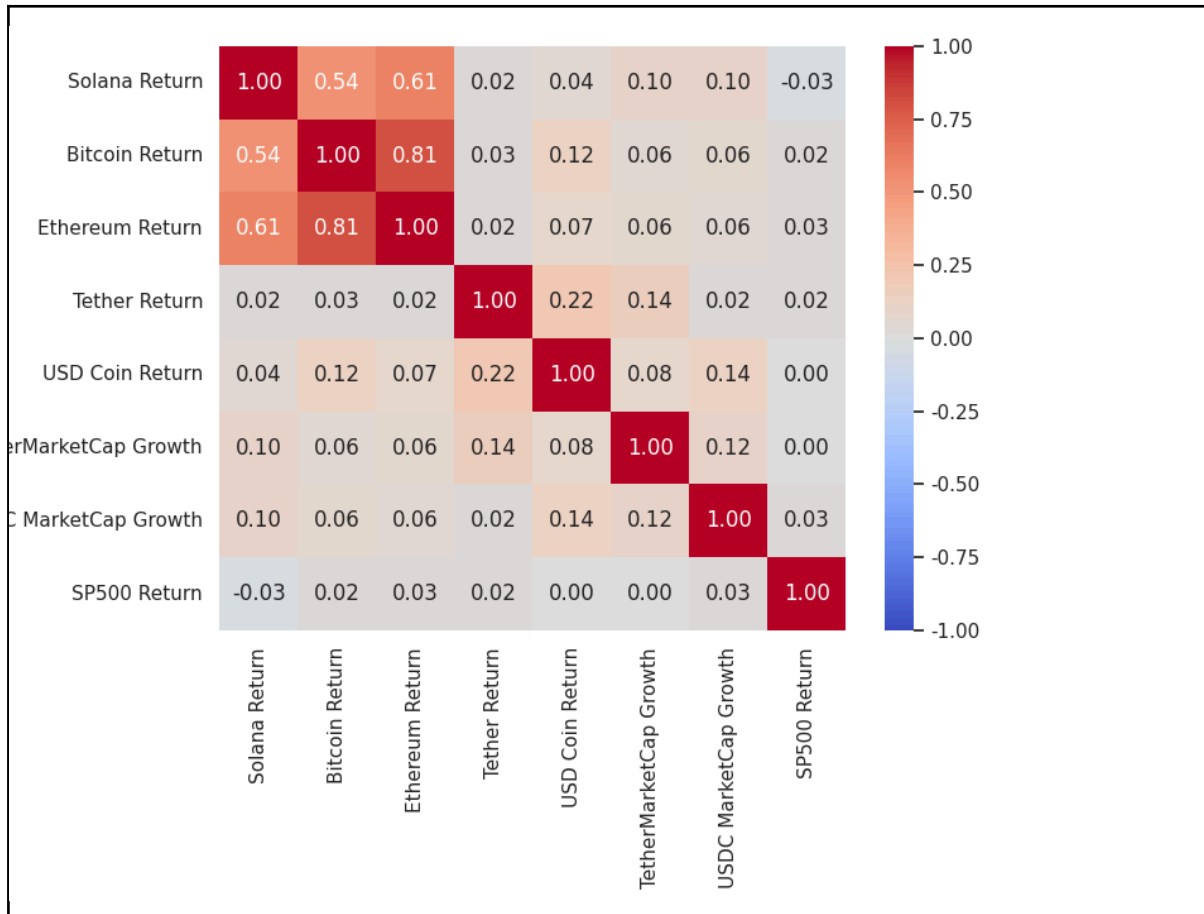
The above Figure 2 and Table 1 use the below data as described from Table 2.

Table 3. Descriptive Statistics

	Solana Returns	Solana Price	Bitcoin Returns	Ethereum Returns	Tether Growth-Rate	USD Coin Growth-Rate	SP500 Returns
Number	1920	1921	1920	1920	1920	1920	1147
Mean	0.27%	78.16	0.15%	0.15%	0.17%	0.24%	0.0005
Standard D	6.55%	71.34	3.12%	4.14%	0.56%	1.05%	0.0106
Minimum	-54.82%	0.51	-17.25%	-30.52%	-3.93%	-7.96%	-0.0608
25%	-3.06%	19.58	-1.22%	-1.76%	-0.02%	-0.19%	-0.0048
50%	0.00%	41.11	0.07%	0.15%	0.07%	0.08%	0.0009
75%	3.48%	143.4	1.53%	2.19%	0.24%	0.50%	0.007
Maximum	38.45%	262.56	17.60%	21.94%	12.23%	10.51%	0.054
Skew	-0.2224	0.62	-0.1033	-0.249	6.9329	1.322	-0.4843
Kurtosis	7.0346	-0.97	3.6233	4.6838	131.5277	17.5656	2.5031

Notes: Volatility for Solana, Bitcoin, and Ethereum is based on daily log returns, while volatility for the stablecoins Tether and USD Coin is derived from daily arithmetic growth rates in market capitalisation. Kurtosis is calculated using Fisher's definition (Pandas, 2024). Data are sourced from CoinGecko and cover the period from 11 June 2020 to 14 July 2025. <https://www.coingecko.com/>

Figure 3. Correlation Matrix of Returns and Market Growth Rates

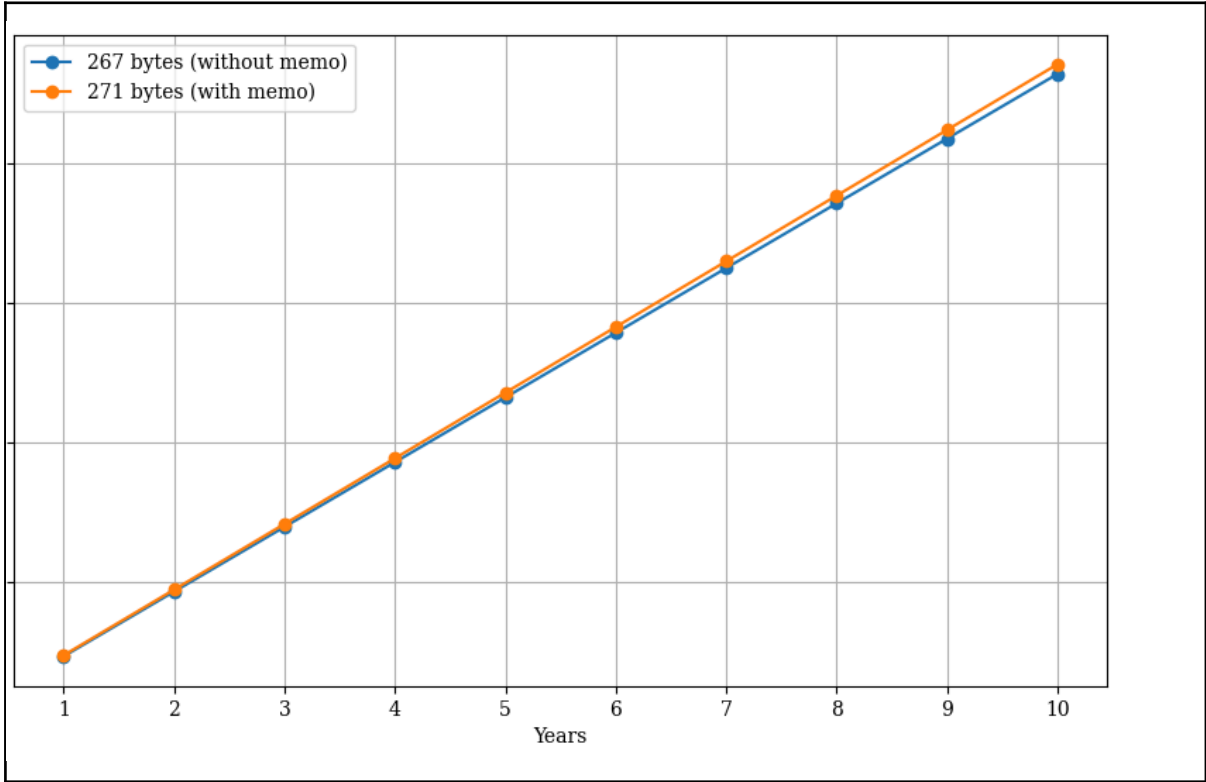


Notes: Cryptocurrency market data were obtained from (<https://www.coingecko.com/>) while S&P 500 price data were retrieved using the Yahoo Finance API. All price series were transformed into log returns, and market capitalisation figures were converted into simple growth rates. The dataset period is subset from 11 June 2020 to 14 July 2025.

Figure 4 illustrates the cumulative rent fees that would accrue on the Solana blockchain when 1,000 transactions are retained on-chain for up to ten years. Two transaction sizes are compared: 267 bytes

(without memo) which is a standard transfer containing only the essential account keys and instructions. And, 271 bytes (with memo), the same transfer augmented with a short memo field. Using Solana’s current rent rate of 0.00000348 SOL per byte-year, the chart shows a linear increase in total rent fees as storage time lengthens. After a decade, the cumulative cost reaches just under 10 SOL for the 267-byte transactions and slightly above 10 SOL for the 271-byte transactions. The near-overlapping lines confirm that adding a small memo has only a marginal impact on long-run storage costs, while the overall slope highlights how rent accrues proportionally with both data size and retention period.

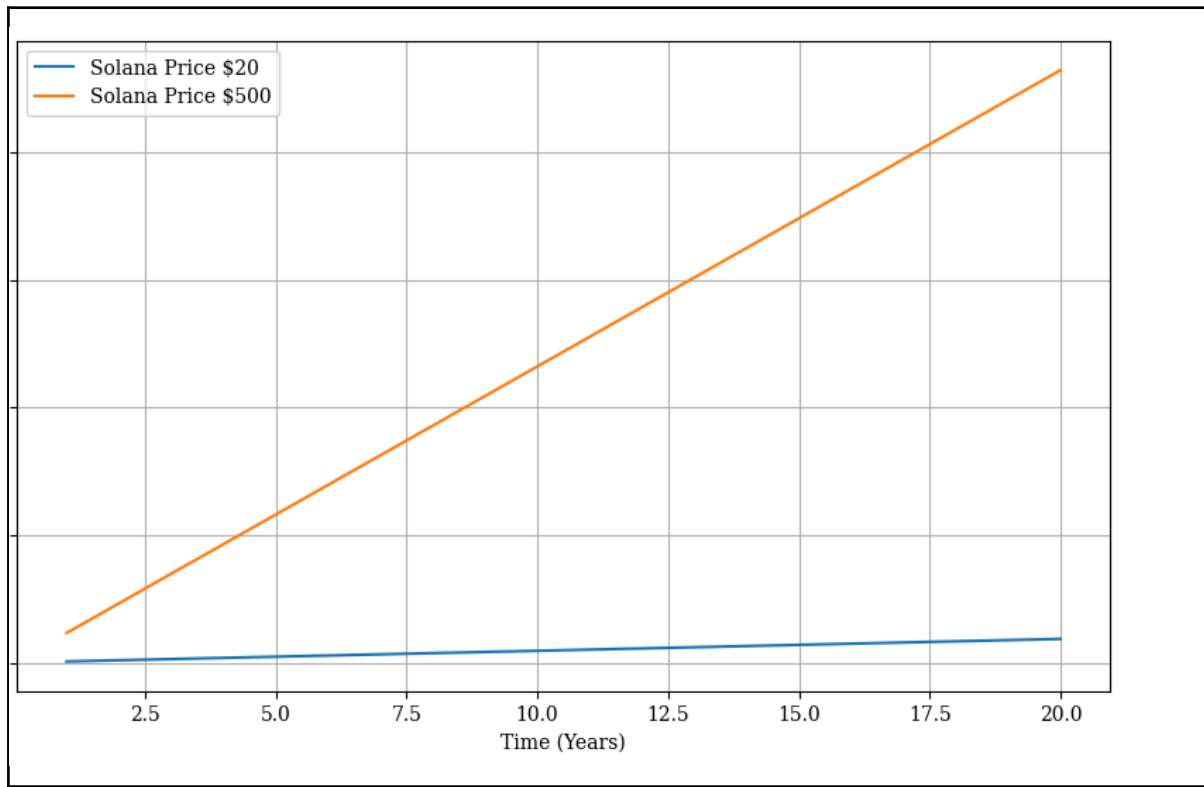
Figure 4. Cumulative Rent Fees Over Time with 1000 Transactions



We may consider a minimum transaction record-keeping period of five years for a one-off payment between parties, and a duration of twenty years for instruments such as bearer bonds. Figure 5 below presents the estimated storage costs for a minimum-sized transaction of 267 bytes over five and twenty years, assuming Solana prices of \$20 and \$500. These price points are evaluated under a normal-distribution assumption; in practice, crypto-asset returns tend to exhibit skewness and fat tails, so the true probability of such extreme values is likely to be higher than indicated by the normal model. In calculating their probabilities from the descriptive statistics in Table 3, the solana price being less than \$20 is around 20% and more than \$500 is less than 0% as demonstrated in the Appendix1 workings¹.

Figure 5. Cumulative Rent Cost for 267 Bytes Contract over 20 Years

¹ To gauge how extreme the illustrative price points of \$20 and \$500 are relative to the historical distribution of daily Solana prices, we assume a normal distribution with sample mean $\mu = 78.16$ and standard deviation $\sigma = 71.34$ (based on CoinGecko data, 11 June 2020 – 14 July 2025). Using the cumulative distribution function package of Scipy in Python, the probability of Solana’s price falling to \$20 or below is approximately 2.7%, while the probability of it reaching \$500 or above is effectively zero ($p \approx 3 \times 10^{-13}$). Because crypto-asset returns display pronounced skewness and excess kurtosis, these normal-based tail probabilities likely understate actual risk or return.



Considering also that there is such a rent-exemption for accounts that reach a certain level of threshold balance. The rent-exemption amount for storing 267 bytes of data on Solana is 0.54448 SOL, which represents the upfront cost needed to make an account rent-exempt, avoiding any rent charges indefinitely. This cost, considering a price of Solana at \$167² can be considered high compared to traditional retail banking. The below graph shows how many contracts can be paid with rent with the threshold balance for rent-exemptions. For instance, for five years storage of 1 transaction in a rent-exempted account would be equal to ~117 rent paid transactions.

Figure 6 plots the break-even number of contracts at which the cumulative rent paid equals the one-off cost of maintaining a rent-exempt balance over holding periods from one to twenty years. At the one-year horizon nearly 600 contracts would be required for rent exemption to be more economical, whereas by five years the threshold falls to roughly 110 contracts. Over a twenty-year period the break-even point drops below 30 contracts, showing that even modest data sets eventually justify the upfront cost of rent exemption. The steep decline in the early years, followed by a gradual flattening, reflects the linear growth of rent charges over time compared with the fixed nature of the rent-exempt deposit, indicating that organisations should compare their expected retention horizon and contract volume with this curve when deciding between paying rent and funding a rent-exempt account.

² Source: Coingecko as of 14th July 2025 <https://www.coingecko.com/>

Figure 6. Break-Even Number of Contracts for Rent Cost vs Rent-Exempt Cost (1 to 20 Years)

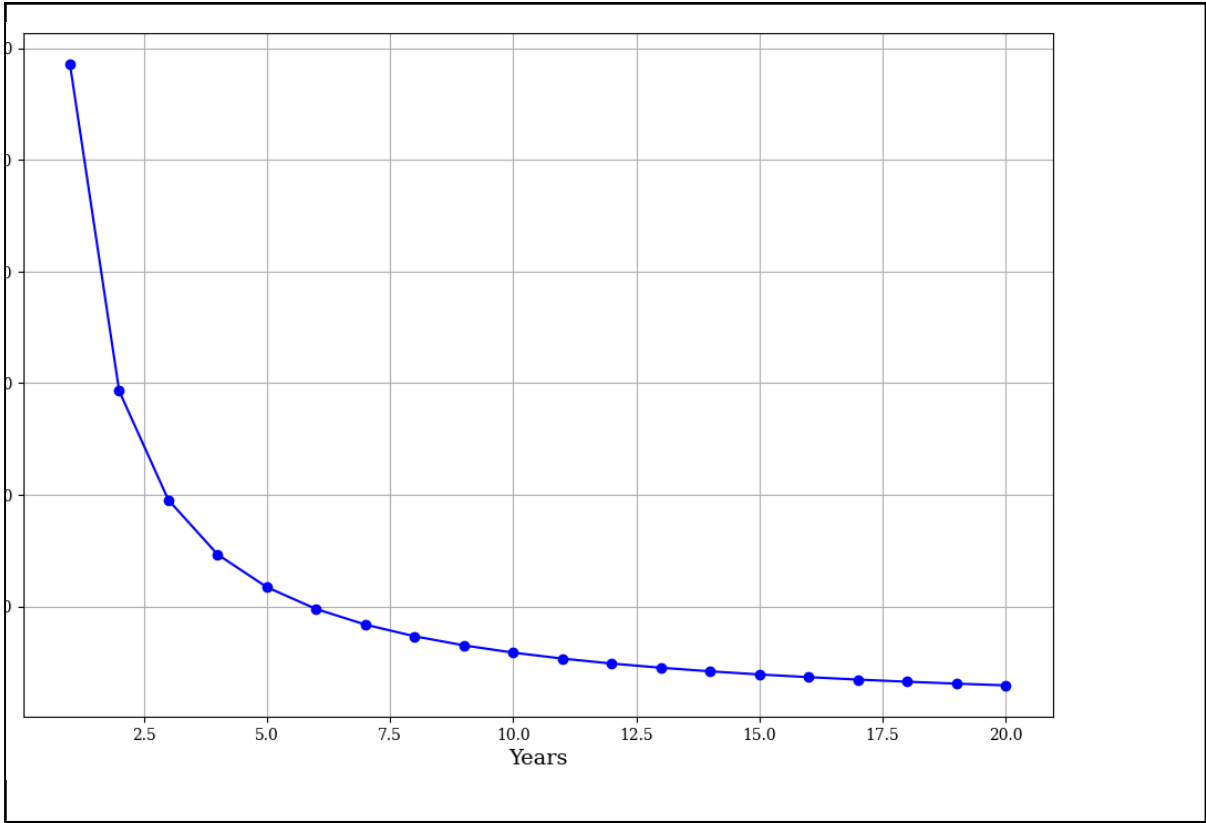


Table 4 presents the same information for the one-, five- and twenty-year horizons as the Figure 6. At one year, the annual rent of 0.000929 SOL per record implies a break-even point of about 586 transactions; at five years, the cumulative rent rises to 0.004646 SOL, reducing the break-even level to 117 transactions; and at twenty years, a total rent of 0.018583 SOL means that storing only 29 transactions would offset the rent-exempt deposit. Together, the figure and the table make clear that rent exemption is economical only when an account is expected to store dozens of records for more than a decade; otherwise, pay-as-you-go rent remains the cheaper option.

Table 4 Rent Exempt

Holding period	Annual rent per 267-byte transaction	Break-even number of transactions*
1 year	0.000929 SOL	$0.54448 \text{ SOL} \div 0.000929 \approx 586 \text{ transactions}$
5 years	$0.000929 \times 5 = 0.004646 \text{ SOL}$	$0.54448 \div 0.004646 \approx 117 \text{ transactions}$
20 years	$0.000929 \times 20 = 0.018583 \text{ SOL}$	$0.54448 \div 0.018583 \approx 29 \text{ transactions}$

Figure 7 and 8 compares five- and twenty-year holding periods. We can see that paying for rent is more cost effective through opportunity cost than holding higher amounts of Solana for rent-exemption.

Figure 7. Comparison of Rent Cost and Rent-Exempt Cost for Storing 267 Bytes per Contract (Five Years)

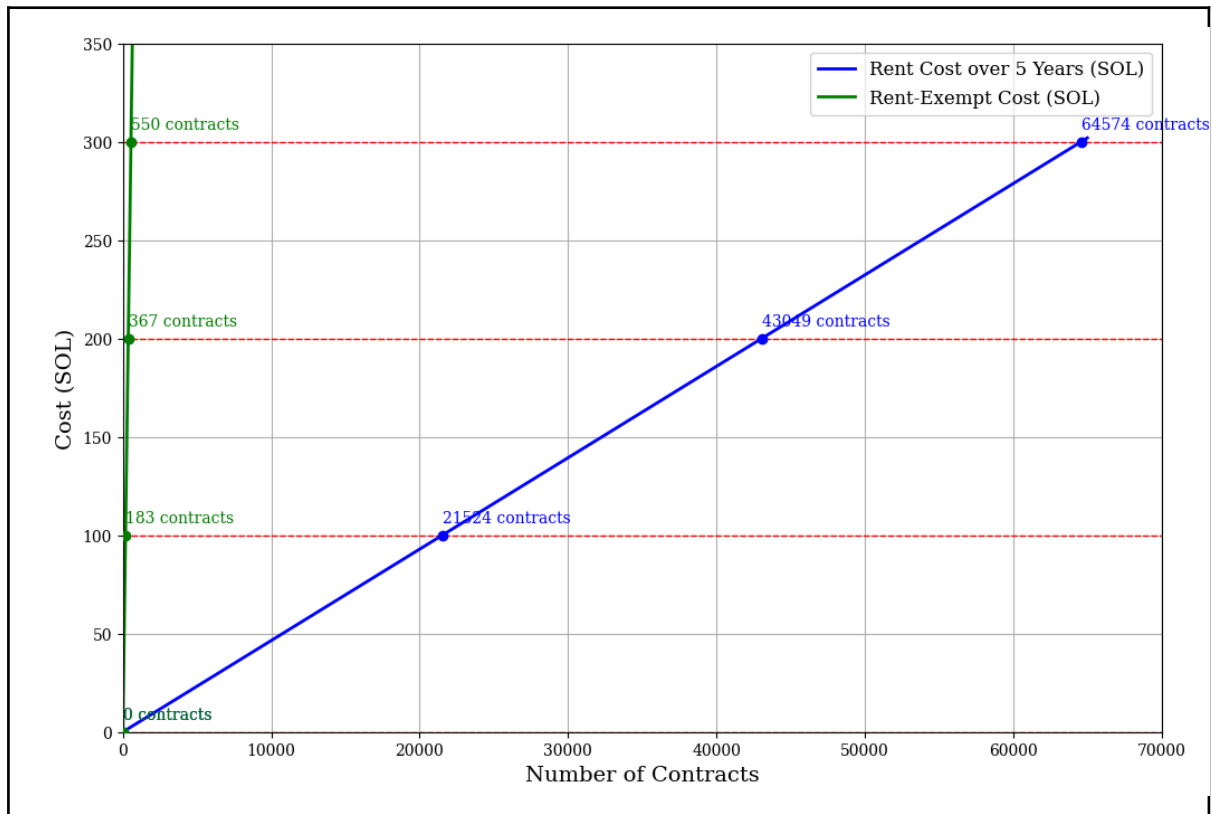
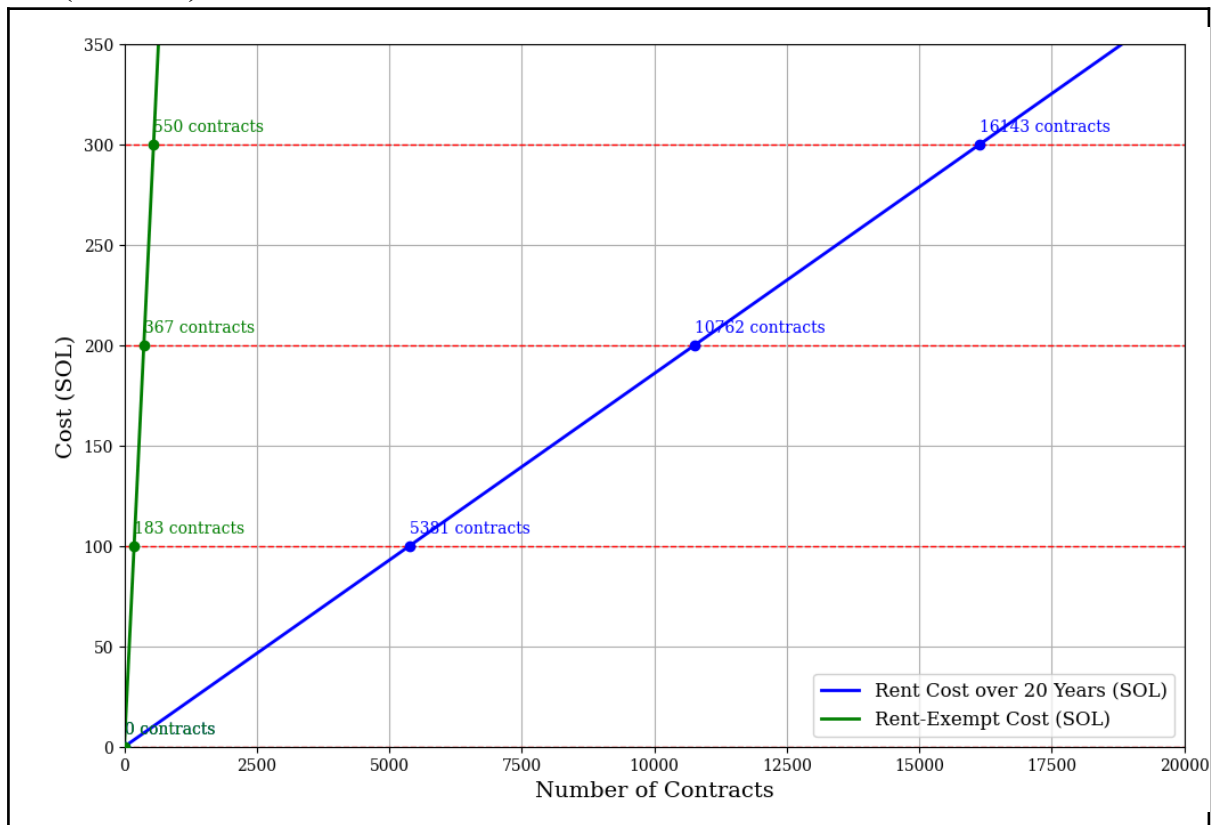


Figure 8. Comparison of Rent Cost and Rent-Exempt Cost for Storing 267 Bytes per Contract (20 Years)



On the basis of the rent analysis, organisations that store large volumes of client transactions will generally find it more economical to pay rent than to fund a rent-exempt balance. A complementary best practice is to maintain a dedicated account for each financial year to record occasional business relationships, while using a separate, ongoing account for clients that trade frequently. This structure

helps ensure compliance with the record-keeping requirements of the Anti-Money-Laundering Directive (AMLD) (EBA, 2024).

A key concern is the reliability of blockchain technology and the level of operational uptime that can be expected. By way of comparison, Microsoft 365 operates under a Service Level Agreement guaranteeing between 99 per cent and 99.99 per cent monthly uptime (Microsoft, 2024). Over the past three years, Solana’s longest recorded outage was a five-hour interruption in February 2024 (Coindesk, 2024a). Microsoft has also experienced disruptions, including a two-hour outage in June 2023 (Reuters, 2023) and a widespread Windows failure in July 2024 that stemmed from a faulty third-party security update (Financial Times, 2024).

An SME can apply game-theoretic reasoning to introduce a complementary feature in a competitive and fast-moving market. By combining blockchain with cloud storage, the firm leverages platform economics (Parker et al., 2016), connecting stakeholders and scaling swiftly by adding new functions, which improves operational efficiency and reduces costs. In this scenario, each batch of oysters is tagged with a QR code whose provenance data is anchored on-chain. Because the information need only be retained for a year and a QR image is far larger than a standard transaction, it is much cheaper to store the bulk of the data off-chain in a cloud database. Recording the QR code entirely on Solana would require a separate on-chain programme and, whether paid through rent or secured via rent exemption, would incur disproportionately high fees (Solana, 2024b). The example therefore shows how a hybrid solution, short-term blockchain anchoring coupled with economical cloud storage, can deliver the necessary integrity while keeping costs under control for an SME.

Example of QR Code Use in Seafood Certification for One Year

For operations built on platform cryptocurrencies, the seafood-certification case provides an instructive example. Hopkins et al. (2024) argue that seafood supply chains require stronger traceability systems. In Nevada, where Las Vegas is based, food establishments may serve molluscan shellfish only if the product is sourced from a dealer that holds a State of Nevada shellfish certificate, issued under the National Shellfish Sanitation Program (Nevada Division of Public and Behavioural Health, 2014). Certification and supply-chain tracking can reduce Information Asymmetry related to the product quality. This notion was listed in the Table 1 related applicable theories. Blockchain anchoring of QR-code references provides a tamper-evident audit trail, while off-chain cloud storage keeps costs low.

To illustrate the cost implications of using Solana for on-chain operations, let’s consider a scenario where a seafood SME tracks oysters with QR codes. Each QR code contains detailed information about the oysters, such as the farm, date, location, and website for contacts or more information. This information can be needed for traceability and compliance with food safety regulations. The assumption is that the farmer prints out the tags with populated information which can then also be read mechanically. We also assume that the records need to be kept for two years.

Cost Calculation for Rent-Exempt Account

The upfront cost needed to make an account rent-exempt on Solana is 0.54448 SOL. Given an example Solana price of \$140 (as of 13.07.2024 - Coingecko), this cost translates to:
 $0.54448 \text{ SOL} \times \$140/\text{SOL} = \$76.23$

Referring to Table 4, the rent-exempt deposit of 0.54448 SOL only becomes cheaper than pay-as-you-go rent once a single account is expected to store more than 586 batch certifications in a single year. Chen et al. (2017) project that a modestly scaled Hawaiian fishpond can market about 156 000 oysters a year. Commercial growers such as Ward Oyster Company in the United States report selling several million oysters annually,

which translates into thousands of batch certificates and is therefore well above the 586-certificate break-even threshold identified in our analysis.

QR Code Storage in Seafood Certification

In the seafood certification example, oysters are tracked using QR codes. The QR code itself doesn't need to store all the information; instead, it contains a link to an external cloud storage where the detailed information is kept. This approach minimises on-chain storage requirements while leveraging blockchain for data integrity and traceability.

On-Chain Storage Cost

Store the QR code link and metadata on Solana. Each transaction to store data on-chain is approximately 267 bytes. The annual rent for 267 bytes is:

$$267 \text{ bytes} \times 0.00000348 \text{ SOL/byte-year} = 0.00092916 \text{ SOL/year}$$

Cost in USD (assuming SOL = \$140 - as of 13.07.2024 - Coingecko):

$$0.00092916 \text{ SOL} \times \$140/\text{SOL} \times 2(\text{years}) = \sim \$0.26$$

Cloud Storage Cost

Assuming each QR code points to data stored in the cloud, with an average cost of \$0.02 per GB per month (common for services like AWS, Google). Data stored per QR code is around 1 KB. The monthly cost for storing 1 KB is \$0.00002.

$$\text{Annual cost for storing 1 KB: } \$0.00002/\text{KB/month} \times 24 \text{ months} = \$0.00048/\text{KB/year}$$

Practical Application and Cost-Benefit Analysis

Using the above calculations for storage costs

On-Chain Cost pay-as-you go rent storage (Solana): \$0.26 for one year.

Cloud Storage Cost: \$0.00048 over two years.

Total pay-as-you go storage cost for tracking each oyster batch with a QR code for one year is:

$$\$0.26(\text{solana rent}) + \$0.00024(\text{cloud}) = \$0.26.$$

On the other hand, under the rent-exemption model, the break-even volume:

$$0.54448 \text{ SOL} \div 0.00185832 \text{ SOL per certificate} \approx 293 \text{ certificates.}$$

With a two-year retention requirement, anchoring a single oyster-batch certificate on Solana and holding the detailed record in the cloud costs roughly \$0.26 in total (about \$0.13 per year) at a Solana price of \$140. By comparison, making the account rent-exempt entails a one-off outlay of \$76.23. At this horizon the rent-exempt deposit becomes cheaper only when an account is expected to store more than approximately 293 certificates in two years, a volume achievable by many mid-sized farms but well above the output of smaller operators. Beyond that point, Economies of Scale, as noted in Table 1, begin to take effect, making it sensible to move away from pay-as-you-go and adopt a rent-exempt or otherwise tailored storage arrangement. Businesses issuing only a few hundred certificates over the two-year period are therefore better served by the pay-as-you-go model combined with low-cost cloud storage; once certificate volumes rise beyond the break-even point, it makes sense to move away from pay-as-you-go and adopt a rent-exempt or otherwise tailored storage arrangement. This goes with the Table 1 included notion of Diminishing Returns. Each additional certificate generates only a small, linear increase in cumulative rent, whereas the rent-exempt deposit is a fixed cost. At low volumes, the marginal savings from switching to rent-exemption is non-existing, so pay-as-you-go remains optimal. Beyond roughly 293 certificates in two years, the marginal benefit of adding further certificates continues to rise but at a decreasing rate, eventually offsetting the upfront deposit.

Beside platform cryptocurrency application as mentioned above, IBM has also developed a

blockchain application to track food distribution for businesses (IBM, 2024). For each net caught of oysters, fisherman stored on the blockchain information such as when the boat landed port-side and when each scallop lot was hand graded, selected, packed and shipped to its final destination (DiCamillo, 2019). Tracking fish is important as mislabeling can have unwanted consequences such as food poisoning. Thus blockchains can help maintain a more efficient supply chain. Beside this, blockchain technology can benefit SMEs by enhancing product traceability, ensuring regulatory compliance, improving operational efficiency, reducing waste, and increasing transparency.

Agency theory (Jensen & Meckling, 1976), highlighted in Table 1, explains how conflicts can arise when one firm in a supply chain acts as an agent on behalf of another firm or the end customer. By anchoring batch certificates on Solana, each party can verify that its upstream partner has recorded provenance data accurately and punctually. The immutable ledger therefore restricts agents' ability to cut corners or withhold information, giving principals greater confidence in product quality and regulatory compliance.

Furthermore, as noting the Asymmetry of Information in Table 1, another emerging feature of blockchain for data storage is that it enables efficient anti-counterfeit checks. High-end goods providers and artisan houses are increasingly utilising blockchain technology to authenticate products and prevent counterfeiting (Ji et al 2023). Such application in the certification business applies also to artworks. For example an identification label or number can be placed on a sculpture or an installation linking to a certificate stored on a chain with information on previous ownerships. This would benefit the auction houses and price consultants, as well as the owners.

A notable instance of blockchain adoption appears in sport, where clubs such as Manchester City and Juventus have launched blockchain-based fan tokens that provide supporters with access to exclusive content, limited-edition merchandise and participation in club polls (Cointelegraph, 2019; 2021). Their purpose is to deepen engagement with, and potentially monetise, the club's fan base, which can be international.

NFTs and tokens therefore offer sports clubs, whether grassroots sides or global brands, a lucrative new revenue stream while deepening fan engagement, as the launches reported by CoinDesk (2024b) illustrate. Clubs venturing into NFTs should pair sales campaigns with clear risk disclosures and robust age-appropriate safeguards, ensuring that commercial innovation is balanced with responsible fan protection. It is important to recognise that digital collectibles are a high-risk asset class. Collectibles generally are a high-risk asset class (Finance Strategists, 2023). Prices can be volatile, secondary-market liquidity is limited and the buyer base often includes unsophisticated investors and minors. Similar to cryptocurrency valuations, NFT valuations are also not derived by traditional financial analysis of discounted cashflows. Behavioural economics helps to explain why these tokens gain rapid traction: supporter communities can be highly susceptible to FOMO, the fear of missing out, and to the endowment effect, in which mere ownership, however notional, confers disproportionate sentimental value (Kahneman, 2011; Thaler and Sunstein, 2008).

Whilst there is a cost of design and integration of the SME proprietary system to the platform cryptocurrency, such as Solana, there is no need to develop proprietary software. In essence, this then also represents a pay as you go model. This represents a Pay-as-you-go Model, which aligns with Disruptive Innovation Theory as listed in Table 1 (Christensen, 1997, 2015), as SMEs can challenge established businesses by adopting scalable technologies without investing in developing proprietary systems. In addition, SMEs can utilise the shared services, often facilitated by a digital platform, and benefit through shared economics before they seek to invest into their bespoke systems (e.g. Knowledge at Wharton, 2015, Lestantri et al. 2022).

As a support feature, the blockchain organisational adaptation has been slow compared to AI or machine learning initiatives. We can compare the US Federal government spend on both of these technologies. Whilst this is not SME data, we can make some guiding insights about the supportive functionalities of

these new tools whilst they are not at the core of the business model. During 2020-2022, federal spending on blockchain technology was less than \$30 million (Deltek, 2023), whereas the investment in AI technologies was in the hundreds of millions of dollars (Larson et al. 2024).

Also, it can be noted that despite the promising applications of blockchain-enabled smart contracts, significant security issues, including vulnerabilities, continue to hinder their widespread adoption (Khan et al. 2021). These potential security vulnerabilities should be carefully evaluated when determining the purpose and selecting the appropriate technology for support services.

SMEs that operate in several jurisdictions often face inconsistent rules on product provenance, environmental reporting and consumer protection. But also, there is the opportunity for wider product or service distribution and also cost optimisation. The SMEs might benefit from Regulatory Arbitrage as noted in Table 1. By anchoring compliance records on Solana etc, a firm can satisfy multiple regulatory regimes with a single, ledger entry instead of maintaining separate databases for each territory. There would need to be access to the Internet. Having supply-chain record keeping on-chain streamlines audits, reduces administrative overheads and avoids the duplication of data-storage infrastructure. Fewer parallel systems mean lower energy consumption and lesser need of system maintenance or set ups, so the blockchain solution not only exploits regulatory gaps to cut costs but also advances sustainability objectives by minimising the firm's operational footprint.

Because their operations are driven by data insights, firms with regional or international reach can spot efficiency gains and new revenue opportunities more quickly than smaller counterparts. This analytical focus, combined with extensive Data Networks as described briefly in Table 1, means they can assess, pilot and scale digital services at a faster pace, placing them among the earliest adopters in the Innovation Diffusion lifecycle as described in Table 1. SMEs can also access these capabilities by using platform-based services and pay-as-you-go technology models. Nevertheless, policy nudges, such as simplified onboarding grants, regulation or interoperability standards, could encourage the late-majority segment of SMEs. While digital infrastructure may not rank as the top investment priority for SMEs, its implementation can yield external advantages including greater operational efficiency and reach.

From a Game Theory perspective, as described in Table 1, these SMEs may also gain a competitive edge by strategically offering additional platform features, such as integrated compliance tools, analytics dashboards, not to increase price but to increase value relative to rivals. By doing so, they influence market expectations and user behaviour, shaping adoption decisions in their favour. However, it is important to note that increased investment in digital tools does not always lead to proportionally greater returns. It's not a linear relationship. Firms should therefore evaluate the appropriate level of investment to avoid over-spending. This ties into the concept of Diminishing Returns, also discussed in Table 1, where additional inputs may yield progressively smaller gains after a breakpoint.

Regulation related to access to platform cryptocurrencies or tax treatments can hinder the adoption by SMEs. The lack of clear regulatory guidelines and the risk of non-compliance with evolving legal frameworks can create uncertainties and additional costs that can deter SMEs from utilising blockchain technologies. Whilst these are global instruments, their SME adaptation heavily depends on regulation (Auer, & Claessens, 2018).

In the United States, federal regulators have taken the position that Bitcoin and now Ether even if it's now Proof-of-Stake can be classified as commodities under the Commodity Exchange Act and therefore fall within the Commodity Futures Trading Commission's (CFTC's) anti-fraud and market-manipulation remit (CFTC, 2018; Behnam, 2023). By contrast, the regulatory status of many alternative tokens remains unsettled. The Securities and Exchange Commission (SEC) has argued in recent enforcement actions that certain proof-of-stake assets, including Solana's native coin, are securities because purchasers allegedly expect profits derived from the efforts of the project's developers (SEC, 2023; Godoy, 2023). No court has yet issued a definitive ruling on SOL's classification, so, for now, it occupies a grey area between the

CFTC's commodity jurisdiction and the SEC's securities oversight. This impacts Solana adoption for SMEs.

Instead, recent legislative milestones are beginning to give stable coins more certain legal foundations on both sides of the Atlantic. In the United States, the Guiding and Establishing National Innovation for U.S. Stablecoins (GENIUS) Act introduces a federal licensing regime for 'payment stablecoin' issuers, classifies fully reserved tokens as non-securities and places prudential supervision with the Office of the Comptroller of the Currency (U.S. Congress, 2025). Within the European Union, the Markets in Crypto-Assets Regulation (MiCA) standardises disclosure, reserve and authorisation requirements for asset-referenced and e-money tokens across all Member States (European Parliament & Council, 2023), while the Distributed-Ledger-Technology Pilot Regime permits regulated venues to trial the issuance and settlement of tokenised shares and bonds in a controlled sandbox (European Parliament & Council, 2022). Collectively, these measures create a more predictable landscape for platforms seeking to securitise assets on-chain, though their ultimate effect on SME finance remains to be assessed.

SMEs stand to benefit from open finance applications built on platform cryptocurrencies, encompassing areas such as payments, financing, customer credit, and accounting functions. However, regulatory frameworks have thus far restricted the adoption of these tools within conventional business models. In response, several banks have begun to develop blockchain-based financial services that comply with existing regulations. For example, Société Générale in France has launched a Euro-denominated stablecoin that is accepted on certain cryptocurrency exchanges (Société Générale-FORGE, 2023; Bitstamp, 2023). These services are generally aimed at institutional and retail users rather than SMEs. More broadly, policymakers—including the European Central Bank—have expressed scepticism towards blockchain and crypto-assets, suggesting that the introduction of central bank digital currencies (CBDCs) may eventually displace cryptocurrencies altogether. Nonetheless, the degree to which CBDCs can serve as substitutes for cryptocurrencies remains an open question in academic discourse (Chiu & Monnet, 2023). As the examples above illustrate, full substitution appears unlikely given the bespoke and programmable nature of many crypto-finance services, which extend beyond payments to support operational processes as well.

Transaction fees on the US credit-card "rails" remain among the highest in the developed world, with merchants typically paying total acceptance costs of around two per cent of the sale value (Nilson Report, 2024). By contrast, dollar-denominated stablecoins and forthcoming retail CBDCs promise near-instant settlement at a fraction of that cost, offering retailers and consumers a lower-fee alternative to traditional card networks. This said, depending on the design of the retail CBDCs, the SMEs could benefit from lower transaction costs using CBDCs. If regulation framework allows, stable coins beside rCBDCs can disintermediate the transaction and reduce costs. This aligns with the concept of Peer-to-Peer Models, as outlined in Table 1. For instance, Swish, a collaborative digital payment system of banks in Sweden, charges a transaction fee of around €0.11 plus 0.60% and an additional SEK 2 (approximately less than €0.20) per transaction (Swish, 2024; Nordea, 2019). In contrast, Visa's fees can range from 1.43% to over 3% depending on the transaction type and card used (Adyen, 2024). In this manner, Swish offers a more cost-effective solution for smaller transactions which benefit SMEs who seek to minimise transaction costs to improve their margins. The economy, through SMEs profitability, would benefit from the stablecoin or rCBDCs introduction as well as the SMEs would have more to invest in their operations.

Although this study concentrates on conventional SME applications of blockchain, the tokenisation of lending, equity and royalties is expanding rapidly and merits its own series of investigations. Roth et al. (2019) show that issuing tokenised shares through equity-crowdfunding platforms can deepen liquidity, widen investor access and reduce dependence on traditional financial service intermediaries. Xu and Vadgama (2021) find that DeFi protocols such as Aave, Compound and Maple-Finance use on-chain

smart contracts to automate peer-to-peer lending, thereby lowering costs of leverage and credit barriers that are typical of traditional finance. While these tools may not yet offer traditional real-economy lending for most SMEs, they can already serve practical purposes in treasury management. Moreaux and Mitre (2022) examine programmable royalty contracts which, once embedded in a blockchain marketplace, distribute payments transparently and in real time to copyright holders. This is essential for providing liquidity for IP stakeholders. These further can be leveraged with companioning these into Agentic AI that can initiate and manage transaction on-chain for financial services or supply-chain operations.

Together, these studies suggest that tokenised finance could reshape the way firms raise capital, manage liabilities and monetise intellectual property. Decentralised autonomous organisations (DAOs) such as Euler Finance extend the concept further by governing pooled capital and development roadmaps through on-chain voting. However, the regulatory, technical and governance complexities of tokenised finance differ markedly from the operational use cases covered here. For that reason, a detailed assessment of tokenised lending, equity and royalties lies beyond the scope of this paper, which remains focused on the practical challenges and opportunities that blockchain creates for SMEs. Although advances in artificial intelligence are expected to lead to the disappearance of certain jobs, these technologies can also lower the barriers to launching or running a business. Platform-based cryptocurrencies are likely to support this transition by providing low-cost, programmable financial and operational tools.

Conclusion

This paper has explored the nuanced application of blockchain technology by SMEs, particularly through the lens of cost minimisation for platform cryptocurrency data storage and transaction verification, using Solana as a case study. In doing so, it has also sought to apply relevant theoretical frameworks that support innovation and can inform strategic decision-making. It has been established that Solana's capability to handle microtransactions with fractions as small as 0.000000001 SOL allows for transaction data storage, enhancing traceability and operational efficiency. This feature not only provides a solution for linking transfer data with internal databases and customer accounts but also significantly reduces external costs associated with traditional data storage methods.

The analysis presented highlights that while the adoption rate of blockchain technology has not matched initial expectations and investment levels, its potential as a complementary service, rather than a core business model, is substantial. The study underscores the importance of understanding the notion of diminishing returns in technology investments, advocating for strategic allocation of resources to technologies that offer the most significant returns relative to their cost and market opportunities.

Moreover, the paper has discussed the economic advantages of utilising platforms and also pay-as-you-go models, which enables SMEs to leverage advanced technologies without substantial upfront investments. These shared economics help to access network effects, competitive service provision and market access. The comparative analysis of blockchain and AI investments by the US Federal government also illustrates the current prioritisation trends, with significantly higher investments in AI.

Additionally, the paper provides a detailed cost analysis of transaction data storage on Solana, emphasising the benefits of paying rent for long-term data storage over holding high amounts of cryptocurrency to achieve a rent-exempt status. This insight is of use when there is a consideration whether data needs to be stored for one year, five year or 20 year periods.

Whilst blockchain technology presents certain challenges, its integration as a support service in financial services and operations offers promising pathways for SMEs to provide cost-effective complementary features. Future research should involve empirical analysis of the state of blockchain tech adoption by SMEs that can be linked to international growth.

References:

- Adyen. (2024, August 21). *The complete guide to credit card processing*. <https://www.adyen.com/knowledge-hub/credit-card-processing>
- Akerlof, G. A. (1970). The market for "lemons": Quality uncertainty and the market mechanism. *The Quarterly Journal of Economics*, 84(3), 488-500.
- Auer, R., & Claessens, S. (2018). Regulating cryptocurrencies: Assessing market reactions. *BIS Quarterly Review*, September 2018, 23 September 2018. Retrieved from https://www.bis.org/publ/qtrpdf/r_qt1809f.pdf
- Behnam, R. (2023, 12 July). Testimony of Chairman Rostin Behnam Before the U.S. Senate Committee on Agriculture, Nutrition and Forestry's Hearing on the Oversight of Digital Commodities. Commodity Futures Trading Commission. <https://www.cftc.gov/PressRoom/SpeechesTestimony/opabehnam48>
- Bitstamp. (2023). Welcome EURCV – Societe Generale – FORGE's Stablecoin. Retrieved from <https://blog.bitstamp.net/post/welcome-eurcv-societe-generale-forges-stablecoin/>
- Chen, J. Q., Haws, M. C., Fong, Q. S. W., & Leung, P. (2017). Economic feasibility of producing oysters using a small-scale Hawaiian fishpond model. *Aquaculture Reports*, 5, 41–51. <https://doi.org/10.1016/j.aqrep.2016.12.001>
- Chiu, J., & Monnet, C. (2023). Money Creation for Distributed Ledgers: Stablecoins, Tokenized Deposits, or Central Bank Digital Currencies?. Retrieved from <https://economics.smu.edu.sg/sites/economics.smu.edu.sg/files/economics/pdf/Seminar/2023/20231208.pdf>
- Christensen, C. M. (1997). *The innovator's dilemma: When new technologies cause great firms to fail*. Harvard Business Review Press.
- Christensen, C. M., Raynor, M., & McDonald, R. (2015). What is disruptive innovation? *Harvard Business Review*, 93(12), 44-53. Retrieved from <https://www.hbs.edu/faculty/Pages/item.aspx?num=50233>
- CoinDesk. (2023). SocGen's Crypto Division, SG Forge, Introduces Euro Stablecoin (EURCV) on Ethereum. Retrieved from <https://www.coindesk.com/business/2023/12/06/socgens-crypto-division-sg-forge-introduces-euro-stablecoin-eurcv-on-ethereum>
- Coindesk. (2024A). Solana network suffers brief outage; SOL steady. Coindesk. Retrieved from <https://www.coindesk.com/markets/2024/02/06/solana-network-suffers-brief-outage-sol-steady/>.
- Coindesk. (2024B). How digital collectibles are shaping athlete legacies. *Consensus Magazine*. Retrieved from <https://www.coindesk.com/consensus-magazine/2024/07/11/how-digital-collectibles-are-shaping-athlete-legacies/>
- CoinGecko (2024), The Fastest Blockchain Processed 91M Transactions in a Day, from <https://www.coingecko.com/research/publications/fastest-blockchains>
- Cointelegraph. (2021, March 19). *Manchester City soccer club launches fan token with Socios*. Cointelegraph. <https://cointelegraph.com/news/manchester-city-soccer-club-launches-fan-token-with-socios>
- Cointelegraph. (2019, December 2). *Juventus soccer club releases blockchain token for fan voting*. <https://cointelegraph.com/news/juventus-soccer-club-releases-blockchain-token-for-fan-voting>
- Commodity Futures Trading Commission. (2018, 3 October). *Federal court finds that virtual currencies are commodities* [Press release No. 7820-18]. <https://www.cftc.gov/PressRoom/PressReleases/7820-18>
- Davidson, S., De Filippi, P., & Potts, J. (2016). Economics of blockchain. Public Choice Conference, Fort Lauderdale, United States. <https://doi.org/10.2139/ssrn.2744751>
- Deltek. (2023). Blockchain technology use and investment in the federal government. GovWin IQ. <https://iq.govwin.com/neo/marketAnalysis/view/Blockchain-Technology-Use-and-Investment-in-the-Federal-Government/7290?researchTypeId=1&researchMarket=PFMAP>
- Dionysopoulos, L., Marra, M., & Urquhart, A. (2025). Stablecoins as digital havens: Cryptocurrencies in conflict and sanctioned economies (FEB-RN Research Paper No. 91/2025). SSRN. <https://doi.org/10.2139/ssrn.5256406>

- DiCamillo, N. (2019). Shellfish plant is putting scallops on IBM's food-tracking blockchain. Coindesk. <https://www.coindesk.com/business/2019/10/17/shellfish-plant-is-putting-scallops-on-ibms-food-tracking-blockchain/>
- Di Maggio, M., Abdallah, R., & Beck, R. (2024). Beyond borders with blockchain - Deloitte's digital vision. Harvard Business School. <https://www.hbs.edu/faculty/Pages/item.aspx?num=224092>
- European Banking Authority (EBA). (2024). Article 40. In Interactive Single Rulebook. Retrieved June 30, 2024, from: <https://www.eba.europa.eu/regulation-and-policy/single-rulebook/interactive-single-rulebook/13119>
- European Parliament and Council. (2022). Regulation (EU) 2022/858 of 30 May 2022 on a pilot regime for market infrastructures based on distributed ledger technology. Official Journal of the European Union, L 151, 1–41. <https://eur-lex.europa.eu/eli/reg/2022/858/oj/eng>
- European Parliament and Council. (2023). Regulation (EU) 2023/1114 of 31 May 2023 on markets in crypto-assets (MiCA). Official Journal of the European Union, L 150, 40–116. <https://eur-lex.europa.eu/eli/reg/2023/1114/oj/eng>
- Financial Times. (2024). Companies around the world hit by Microsoft outage. <https://www.ft.com/content/fba9b61d-efcf-4348-b640-ccb1f9d18ced>
- Finance Strategists. (2023). Collectibles investments: Meaning, types, risks, & strategies. <https://www.financestrategists.com/articles/collectibles-investments/>
- Godoy, J. (2023, June 7). *What makes a crypto asset a security in the U.S.?* Reuters. <https://www.reuters.com/business/finance/what-makes-crypto-asset-security-us-2023-06-07>
- Hawlitschek, F., Notheisen, B., & Teubner, T. (2018). The limits of trust-free systems: A literature review on blockchain technology and trust in the sharing economy. *Electronic Commerce Research and Applications*, 29, 50-63. <https://doi.org/10.1016/j.elerap.2018.03.005>
- Hopkins, C. R., Roberts, S. I., Caveen, A. J., Graham, C., & Burns, N. M. (2024). Improved traceability in seafood supply chains is achievable by minimising vulnerable nodes in processing and distribution networks. *Marine Policy*, 159, 105910. <https://doi.org/10.1016/j.marpol.2023.105910>
- Houston, J. F., Lin, C., & Ma, Y. (2012). Regulatory arbitrage and international bank flows. *Journal of Finance*, 67(5), 1845-1895. <https://www.jstor.org/stable/23324385>
- IBM. (2024). IBM Supply Chain Intelligence Suite: Food Trust. Retrieved from <https://www.ibm.com/products/supply-chain-intelligence-suite/food-trust>
- Ji, T., Goyal, S.B., Ibrahim, A.Z.M. (2023). Blockchain Technology-Based Framework for Anti-Counterfeiting and Traceability. In: Swaroop, A., Kansal, V., Fortino, G., Hassanien, A.E. (eds) *Proceedings of Fourth Doctoral Symposium on Computational Intelligence*. DoSCI 2023. Lecture Notes in Networks and Systems, vol 726. Springer, Singapore. https://doi.org/10.1007/978-981-99-3716-5_14
- Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.
- Karkkainen, T., & Broussard, J. P. (2024). Stablecoin market growth and sensitivity to interbank lending rates: Projections for wholesale CBDC adoption (SSRN Working Paper No. 4864176). SSRN. <https://papers.ssrn.com/abstract=4864176>
- Khan, S. N., Loukil, F., Ghedira-Guegan, C., Benkhelifa, E., & Bani-Hani, A. (2021). Blockchain smart contracts: Applications, challenges, and future trends. *Peer-to-Peer Networking and Applications*, 14(5), 2901-2925. <https://doi.org/10.1007/s12083-021-01127-0>
- Knowledge at Wharton. (2015). *The Sharing Economy: What Is a Sharing Economy?* Retrieved from <https://knowledge.wharton.upenn.edu/article/the-sharing-economy-a-new-way-of-doing-business/>
- Larson, J., Denford, J. S., Dawson, G. S., & Desouza, K. C. (2024). The evolution of artificial intelligence (AI) spending by the U.S. government. Brookings Institution. Retrieved from <https://www.brookings.edu/articles/the-evolution-of-artificial-intelligence-ai-spending-by-the-u-s-government/>
- Lestantri, I. D., Janom, N. B., Aris, R. S., & Husni, Y. (2022). The perceptions towards the digital sharing economy among SMEs: Preliminary findings. *Procedia Computer Science*, 197, 82-91. <https://doi.org/10.1016/j.procs.2021.12.121>

Malthus, T. R. (1798). An essay on the principle of population. J. Johnson.

Marshall, A. (1920). Principles of economics (8th ed.). Macmillan.

Metcalfé, B. (2013). Metcalfe's law after 40 years of Ethernet. IEEE Computer. 46 (12): 26–31.

Microsoft. (2024). Service Level Agreement for Microsoft Online Services. (June, 2024) Retrieved 16.07.2024 from <https://www.microsoft.com/licensing/docs/view/Service-Level-Agreements-SLA-for-Online-Services>

Moreaux, A. C., & Mitrea, M. P. (2023). Royalty-friendly digital asset exchanges on blockchains. IEEE Access, 11. <https://scispace.com/pdf/royalty-friendly-digital-asset-exchanges-on-blockchains-25h2ii6x.pdf>

Nevada Division of Public and Behavioural Health.(2014). Information for a shellfish certificate. Factsheet. https://dphh.nv.gov/uploadedFiles/dphh.nv.gov/content/Reg/Regulatory/Docs/Web_Shellfish_Information.pdf

The Nilson Report. (2025, March). Merchant processing fees in the United States – 2024 (Issue 1281, pp. 1–3). The Nilson Report. Retrieved 20 July 2025, from <https://nilsonreport.com/articles/merchant-processing-fees-in-the-united-states-2024/>

Nordea. (2019). The benefits of Swish for businesses in Sweden. <https://www.nordea.com/en/news/the-benefits-of-swish-for-businesses-in-sweden>

OECD. (2019). Initial Coin Offerings (ICOs) for SME Financing. <https://search.oecd.org/finance/initial-coin-offerings-for-sme-financing.htm>

Pandas Development Team. (2025). pandas.DataFrame.kurt. Pandas documentation. <https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.kurt.html>

Parker, G. G., Van Alstyne, M. W., & Choudary, S. P. (2016). Platform revolution: How networked markets are transforming the economy and how to make them work for you. W.W. Norton & Company.

Polkadot. (2025). DOT token. Polkadot Wiki. <https://wiki.polkadot.network/learn/learn-dot/>

Reuters. (2023). Microsoft 365 down for thousands of users. Retrieved from <https://www.reuters.com/technology/microsoft-365-down-thousands-users-downdetector-2023-06-05/>

Ricardo, D. (1817). Principles of political economy and taxation. John Murray.

Rogers, E. M. (2003). Diffusion of innovations (5th ed.). Free Press.

Romer, P. (2007). Economic growth. Library of Economics and Liberty. Retrieved from <https://www.econlib.org/library/Columns/y2007/Romergrowth.html>

Roth, J., Schär, F., & Schöpfer, A. (2019). The tokenization of assets: Using blockchains for equity crowdfunding. Center for Innovative Finance, University of Basel. SSRN. <https://doi.org/10.2139/ssrn.3443382>

Smith, A. (1776). The wealth of nations. W. Strahan and T. Cadell.

Societe Generale-FORGE. (2023). CoinVertible: The First Institutional Stablecoin Deployed on a Public Blockchain. Retrieved from <https://www.sgforge.com>

Solana. (2024A). Fees on Solana. Retrieved June 16, 2024, from <https://solana.com/docs/core/fees>

Solana. (2024B). Deploying programs. Solana Documentation. <https://solana.com/docs/programs/deploying>

Solana Explorer. (2024). Solana Chain Explorer. Retrieved 7.07.2024 from <https://explorer.solana.com/>

Stiglitz, J. E. (2002). Information and the change in the paradigm in economics. The American Economic Review, 92(3), 460-501.

Swish. (2024.). About Swish. Retrieved from <https://www.swish.nu/about-swish>

Thaler, R. H., & Sunstein, C. R. (2008). Nudge: Improving decisions about health, wealth, and happiness. Yale University Press.

- Tian, F. (2017). A supply chain traceability system for food safety based on HACCP, blockchain & Internet of things. In 2017 International Conference on Service Systems and Service Management (pp. 1-6). Dalian, China. <https://doi.org/10.1109/ICSSSM.2017.7996119>
- U.S. Securities and Exchange Commission. (2023). Complaint, SEC v. Coinbase Global, Inc., No. 1:23-cv-04738 (S.D.N.Y. filed 6 June 2023). <https://www.sec.gov/enforcement-litigation/litigation-releases/lr-25751>
- United States Congress. (2025). Guiding and Establishing National Innovation for U.S. Stablecoins Act (GENIUS Act), Pub. L. No. 118-214. <https://www.congress.gov/bill/119th-congress/senate-bill/394/text>
- Varian, H. R., & Shapiro, C. (1998). Information rules: A strategic guide to the network economy. Harvard Business School Press.
- von Neumann, J., & Morgenstern, O. (1944). Theory of games and economic behavior. Princeton University Press.
- Wang, Q., Li, R., Wang, Q., & Chen, S. (2021). Non-fungible token (NFT): Overview, evaluation, opportunities and challenges. *arXiv*. <https://doi.org/10.48550/arXiv.2105.07447>
- WSJ.(2025, June 25). The Big loser from the GENIUS Act is \$156 billion crypto giant Tether. The Wall Street Journal. <https://www.wsj.com/finance/currencies/tether-stablecoin-genius-act-e403a59b>
- Yermack, D. (2017). Corporate governance and blockchains. *Review of Finance*, 21(1), 7–31. <https://doi.org/10.1093/rof/rfw074>
- Xu, J., & Vadgama, N. (2022). From banks to DeFi: The evolution of the lending market. In *Enabling the Internet of Value* (pp. 53–66). Springer. https://doi.org/10.1007/978-3-030-78184-2_6